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Viewpoint

CO_2 emissions from electricity generation in seven Asia-Pacific and North American countries: A decomposition analysis

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ABSTRACT

The Logarithmic Mean Divisia Index (LMDI) method of complete decomposition is used to examine the role of three factors (electricity production, electricity generation structure and energy intensity of electricity generation) affecting the evolution of CO_2 emissions from electricity generation in seven countries. These seven countries together generated 58% of global electricity and they are responsible for more than two-thirds of global CO_2 emissions from electricity generation in 2005. The analysis shows production effect as the major factor responsible for rise in CO_2 emissions during the period 1990–2005. The generation structure effect also contributed in CO_2 emissions increase, although at a slower rate. In contrary, the energy intensity effect is responsible for modest reduction in CO_2 emissions during this period. Over the 2005–2030 period, production effect remains the key factor responsible for increase in emissions and energy intensity effect is responsible for decrease in emissions. Unlike in the past, generation structure effect contributes significant decrease in emissions. However, the degree of influence of these factors affecting changes in CO_2 emissions vary from country to country. The analysis also shows that there is a potential of efficiency improvement of fossil-fuel-fired power plants and its associated co-benefits among these countries.

1. Introduction

There is growing scientific evidence of an increase in greenhouse gas (GHG) concentrations in the atmosphere since pre-industrial times contributing to rising global temperatures and to changes in climate patterns. The primary source of these increased atmospheric GHG concentrations has been the rising fossil-fuel use and other human activities. Since 1750, it is estimated that about two-thirds of anthropogenic CO_2 emissions—the most important anthropogenic GHG—have come from fossil-fuel burning and in recent years these emissions have continued to increase.

Power generation—which includes both electricity and heat generation—is one of the major sources of CO_2 emissions from fossil-fuel combustion. For example, the share of power generation in global energy-related CO_2 emissions has increased from 36% (8.8 Gt CO_2) in 1990 to 41% (11.0 Gt CO_2) in 2005 and if the current trends continues this share is projected to increase to 45% (18.7 Gt CO_2) in 2030 (IEA, 2007a). Therefore implications of power generation on climate-change mitigation have become an increasingly important topic for researchers and policymakers. However, establishing any effective policies and measures to reduce CO_2 emissions from this sector requires understanding of trends and factors affecting these emissions. Many factors influence these emissions, including socio-economic developments, structural (fuel-mix) change, technological change and institutional frameworks. Few studies have attempted to address some of these issues in the past. For example, Shrestha and Timilsina (1996) studied these factors in power sector for several Asian countries covering fossil fuels only, while Ang et al. (1998) and Ang and Choi (2002) studied these factors in power sector for South Korea. In this study, these factors affecting the changes in CO_2 emissions explicitly from electricity generation over the period 1990–2005 are analyzed for seven major Asia-Pacific and North American countries: Australia, Canada, China, India, Japan, South Korea (hereafter Korea) and the United States (US).[1] Besides, the selection of countries also reflects the differences in their economic development and energy use patterns. These seven countries (hereafter APP countries) together represent about half of the world's economy, population and energy use (UN, 2006; IEA, 2007b, c). In 1990, APP countries together emitted over

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[1] These seven countries are also the partner countries of the Asia-Pacific Partnership on Clean Development and Climate (APP) established in 2006 to address increased energy needs and the associated issues of air pollution, energy security, and climate change between the member countries.

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two-thirds of global energy-related CO_2 emissions and this share has increased to 78% in 2005 (IEA, 2007d). Excluding Australia, these six countries are also the top 10 world CO_2 emitting countries in 2005.

In recent years, decomposition analysis technique provides widely used analytical tools for studying the factors affecting changes in energy and gas emissions. There are two commonly used decomposition analysis techniques in the literature: Index Decomposition Analysis (IDA) that uses aggregate data at the sectoral level and Structural Decomposition Analysis (SDA) that uses input–output tables. Among different IDA techniques, Ang (2004) recommended Logarithmic Mean Divisia Index (LMDI) method over other IDA methods because this method has several desirable advantages including time independence, ability to handle zero values and consistency in aggregation. Few recent studies (Chung and Rhee, 2001; Lenzen, 2006; Rhee and Chung, 2006; Wood and Lenzen, 2006) raised the issues of negative values and large number of zero values in the data set in the application of LMDI method. However, an analysis by Ang and Liu (2007a) shows that replacing zero values with sufficiently small positive value (e.g., δ^{-150}) would lead to satisfactory decomposition results. In the case of negative values, Ang and Liu (2007b) proposed a procedure to deal with negative values for the LMDI method. In this paper, the IDA technique using additive LMDI method is adopted.

2. Overview of electricity generation and CO_2 emissions in APP countries

Over the past 15 years, total electricity generation in APP countries as a whole grew by average 3.7% per year compared to 3.2% per year worldwide, 1.5% per year in OECD Europe as a whole and 6.8% in Developing Asia (DA) as a whole (Table 1). Between 1990 and 2005, three countries—China (9.7% per year), Korea (8.8% per year) and India (6.1% per year)—experienced the fastest rate of electricity generation growth while this growth is modest in industrialized countries ranging from 1.7% per year in Canada to 2.8% per year in Australia. It is noteworthy that three countries (China, Japan and the US) combined accounted for 80% of APP countries combined and 46% of global electricity generation in 2005. With the exception of Japan, the share of fossil-fuel-fired power plants in total electricity generation in six countries has increased over the past 15 years with Korea and India showing the largest increase. Although the share of coal use in total electricity generation has increased between 1990 and 2005, the reason for decline in Japan's fossil-fuel share is mainly due to increase in nuclear share and decrease in oil share in total electricity generation. In 2005, about 82% of total electricity in China and India is generated from fossil-fuel-fired power plants—which is higher compared to world average (67%) and OECD Europe average (53%)—followed by the US (73%), Australia (67%), Japan (63%), Korea (61%) and Canada (26%). The reason for relatively low share in Canada is mainly due to dominant shares of hydro (58%) and nuclear (15%) power plants in total electricity generation. Coal use dominated in total fossil-fuel-based electricity generation in APP countries ranging from 45% in Japan to almost 97% in China in 2005. Between 1990 and 2005, oil use in total electricity generation however continues to decline in six countries except in India, while gas use continues to increase in all APP countries.

The electricity sector has been the main source of total CO_2 emissions in APP countries for decades. The share of electricity generation of APP countries together in total global energy-related CO_2 emissions increased from 61% in 1990 to almost 68% in 2005. The sector's CO_2 emissions in APP countries combined in 2005 is more than six-fold than that of OECD Europe combined and almost twelve-fold than that of DA as a whole. Between 1990 and 2005, Korea (448%), China (301%) and India (169%) lead the three digit percentage increase in emissions. However, in absolute values, the US and China are by far the two biggest emitters both in 1990 and 2005 while India and Japan are the two second biggest emitters. The US and China combined accounted for more than 73% of total APP countries and almost half of the global CO_2 emissions from electricity generation in 2005 (Table 1). Australia, Canada and Korea combined however emitted only about 8% of total APP countries emissions in 2005.

From 1990 to 2005, average efficiency of fossil-fuel-fired power plants has increased in all APP countries. However, at the plant level, some of these efficiencies have decreased. For example, average efficiency of coal-fired power plants has decreased in India and in the US between 1990 and 2005. Likewise, average efficiency of oil-fired power plants has decreased in the US while

Table 1
Overview of electricity sector and corresponding CO_2 emissions in APP countries and selected world regions in 1990 and 2005

Country/region	Total electricity generation[a] (TWh)		Share of fossil-fuel-fired electricity generation (%)		CO_2 emissions from electricity generation[b] (Mt CO_2)		Average efficiency of fossil-fuel-fired power plants (%)					
							Coal		Oil		Gas	
	1990	2005	1990	2005	1990	2005	1990	2005	1990	2005	1990	2005
Australia	153	234	91	93	125	204	35.6	35.6	41.5	39.2	40.5	30.7
Canada	482	616	22	24	95	119	35.4	38.6	38.0	41.3	49.6	42.2
China	621	2497	80	82	552	2115	28.9	32.1	34.0	34.1	38.9	38.9
India	289	699	73	82	236	639	28.4	26.5	24.0	32.7	24.7	41.9
Japan	837	1094	64	63	361	464	39.2	42.2	44.0	47.9	42.9	45.2
Korea	99	353	40	57	34	145	25.9	39.3	37.9	43.3	40.5	52.5
USA	2999	3936	69	71	1734	2234	37.2	36.8	39.5	36.8	37.2	45.7
APP total	5480	9430	66	71	3136	5921	34.7	34.1	40.4	39.8	38.9	45.0
DA[c]	331	895	64	79	180	504	31.6	32.9	34.3	38.8	33.2	41.6
OECD-E[d]	1126	1399	95	91	947	918	36.0	38.9	39.0	35.3	40.8	51.4
World	10131	16338	58	64	5150	8742	34.8	34.6	36.9	36.7	36.2	41.4

Sources: IPCC (2006), IEA (2007b–d).

[a] Excludes electricity generation from combined heat and power (CHP) plants.

[b] Excludes emission from industrial and non-renewable municipal wastes.

[c] Developing Asia (DA) excluding China and India.

[d] OECD Europe. Regional definitions of DA and OECD Europe are based on International Energy Agency classification.

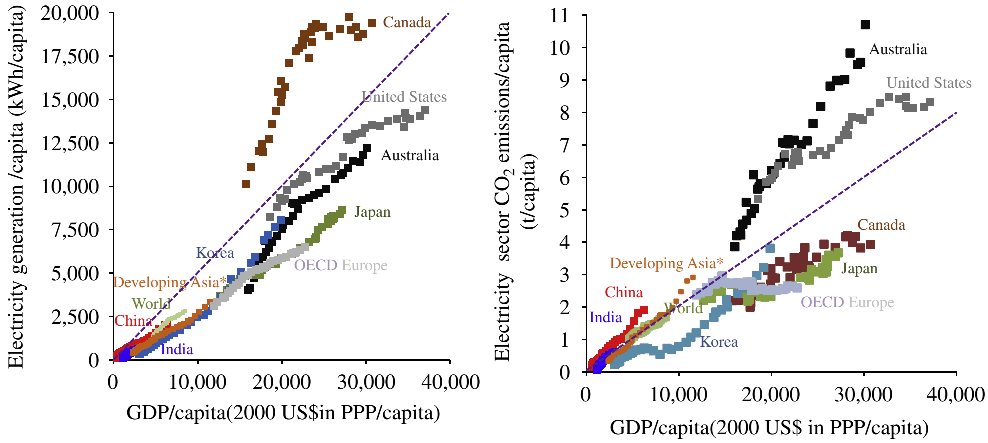

Fig. 1. Relation between income and electricity generation (left) and relation between income and CO_2 emissions from electricity generation (right) in per capita terms for APP countries and selected world regions for the period 1971–2005. *Note*: *Excludes China and India. *Sources*: UN (2006), IEA (2006a), IEA (2007b–d).

average efficiencies of oil- and gas-fired power plants have decreased in Australia during the same period. But this decrease in average efficiencies—which are calculated using IEA data set—need to be looked carefully.[2] The average efficiency of existing fossil-fuel-fired power plants varies from country to country. The efficiency of coal-fired plants ranges from about 26.5% in India to 42.2% in Japan in 2005. Across the countries, Korea shows strong increase in efficiency of coal-fired plants: 25.9% in 1990 to 39.3% in 2005. In 2005, average efficiency of oil-fired plants ranges from 32.7% in India to 47.9% in Japan while average efficiency ranges from 30.7% in Australia to 52.5% in Korea for gas-fired plants. The efficiency of gas-fired plants in four countries (Australia, Canada, China and India) is below 45% while it is over 45% in three countries (Japan, Korea and the US) in 2005.

High levels of electricity generation have close correlation with high levels of income. As shown in Fig. 1 (left), both per capita gross domestic product (GDP) and per capita electricity generation in developing countries of Asia (i.e., China, India and DA average) are much lower compared to industrialized nations and the OECD Europe average over the 1971–2005 period. However, as developing countries become more prosperous, electricity generation is also expected to increase significantly. For example, between 1971 and 2005, average annual per capita electricity generation increased by 25.1% in Korea, 11.4% in China and 5.4% in India compared to only 3.0% in Australia, 2.4% in Japan, 1.9% in Canada and 1.8% in the US. It is projected that per capita electricity generation in China and India continue to grow strong at average 3.0% per year in the next 25 years due to strong economic growth. The trend line shown in Fig. 1 (left) illustrates roughly the average amount of electricity generation per capita relative to GDP per capita. Reasons for high level of electricity generation per capita relative to GDP per capita in Canada include low population density and a relatively cold climate.

Similarly, there is a strong correlation between income and electricity sector CO_2 emissions in per capita terms. The trend line shown in Fig. 1 (right) illustrates roughly the average amount of electricity sector CO_2 emissions per capita relative to GDP per capita. On average, countries and regions above this line are more carbon intensive in electricity generation (e.g., Australia and the US) compared to those below the line. The industrialized nations have higher levels of CO_2 emissions per capita, in proportion to their higher GDP per capita. Including OECD Europe average, two countries (Canada and Japan) are below the trend line while Korea is moving towards the trend line in recent years. Several factors can influence the position of a country or region relative to the trend line including socio-economic development, energy efficiency and fuel mix. Of the industrialized countries, e.g., Australia has a low population density and relies heavily on coal for its electricity generation. The US has low population density in comparison with OECD Europe average and Japan, and the US relies more than two-thirds of its electricity generation from fossil fuels. Japan, Korea and OECD European countries have relatively dense populations and they have nuclear power generation capacity. Factors contributing to Canada's position include a low population density and relies heavily on renewable (mainly hydroelectricity) for its electricity generation. Of the developing countries of Asia (i.e., China, India and DA average), they are situated slightly above the trend line particularly in recent years. Although these developing Asian countries have relatively high population density compared to industrialized nations, results of rapid economic growth, heavy reliance on coal for electricity generation and lower average electricity generation efficiency lead these countries to be situated above the trend line.

3. Methodology

Following the IPCC (2006) method, the total CO_2 emissions from electricity generation (C) is estimated by

$$C = \sum_i C_i = \sum_i FC_i \times EF_i \times O_i, \quad i = 1, 2, 3, \ldots .i, \ldots, n, \tag{1}$$

where C_i is the CO_2 emissions from electricity generation from energy source i, FC_i is the amount of energy i used, EF_i is the CO_2 emission factor of energy source i and O_i is the fraction of carbon oxidized of energy source i.

The IDA identity for evaluating CO_2 emissions from electricity generation is given by

$$C = \sum_i C_i = \sum_i G \frac{G_i}{G} \frac{Q_i}{P_i} \frac{C_i}{Q_i} = \sum_i G s_i f_i u_i, \tag{2}$$

[2] A study by Graus et al. (2007) reported that Indian IEA statistics show 10–12% higher coal input when compared with Indian national statistics for the period 1990–2003 mainly due to differences in conversion factors. In the US, their study reported that IEA statistics of the US is based on gross electricity generation while the US's Energy Information Administration (EIA) statistics are based on net electricity generation. In the case of Australia, oil-fired electricity generation is very small only about 1.9 TWh in 2005 (i.e., 0.8% of total generation); while the reason for decrease in efficiency of gas-fired electricity generation between 1990 and 2005—which generated about 22 TWh in 2005 (i.e., 9.5% of total generation)—is not identified. However, throughout the study, IEA energy statistics are used to maintain the consistency in comparison across the countries.

where G and G_i are the total electricity generation and generation from energy source i, respectively, Q_i is the energy input to electricity generation using energy source i, $s_i = G_i/G$ is the electricity generation share of energy source i, $f_i = Q_i/P_i$ is the energy generation intensity of energy source i and $u_i = C_i/Q_i$ is the CO_2 emission factor of energy source i. The energy source i includes both fossil fuels and non-fossil fuels. Only fossil fuels emit CO_2 emissions.

Following the complete additive LMDI decomposition without any residual term proposed by Ang (2004) and manipulating Eq. (2), the aggregate change in CO_2 emissions level from year t to year t+1 (ΔC_{tot}) is expressed as

$$\Delta C_{tot} = \Delta C_{pdn} + \Delta C_{gmx} + \Delta C_{ein} + \Delta C_{emf}, \tag{3}$$

where ΔC_{pdn} is the change in total electricity production effect, ΔC_{gmx} is the change in electricity generation mix effect, ΔC_{ein} is the change in energy generation intensity effect and ΔC_{emf} is the change in emission factor effect. In Eq. (3), the production effect (hereafter *PDN*) represents the change in CO_2 emissions due to total electricity generation, the electricity generation mix effect (hereafter *GMX*) shows the change in CO_2 emissions due to a structure of generation share of different types of power plants in total electricity generation (hereafter *EIN*) and the emission factor effect captures the change of CO_2 emissions due to changes in emission factor. As the CO_2 emission factor for each of the fossil fuels is assumed to remain unchanged over the study period, $\Delta C_{emf} = 0$ in Eq. (3).The additive LMDI formulae used in the study for three decomposition factors between year t and t+1 are:

$$\Delta C_{pdn} = \sum_i L(C_i^{t+1}, C_i^t) \ln\left(\frac{G^{t+1}}{G^t}\right),$$

$$\Delta C_{gmx} = \sum_i L(C_i^{t+1}, C_i^t) \ln\left(\frac{s_i^{t+1}}{s_i^t}\right)$$

and

$$\Delta C_{ein} = \sum_i L(C_i^{t+1}, C_i^t) \ln\left(\frac{f_i^{t+1}}{f_i^t}\right), \tag{4}$$

where $L(C_i^{t+1},\ C_i^t) = (C_i^{t+1} - C_i^t)/(\ln C_i^{t+1} - \ln C_i^t)$ is the logarithmic mean function.

4. Data

In this study, data from several publications from IEA are used. Historical data (1971–2005) on energy input to electricity generation, electricity generation output by each energy sources and the aggregate real GDP in purchasing power parity (PPP) are based on IEA (2007b, c). Future projected data for these variables are based on IEA (2007a). Historical population which is used for calculating per capita indices shown in Fig. 1 is based on World Population Prospects of UN (UN, 2006). For estimating CO_2 emissions from electricity generation, data on emission factors is based on default emission factors for stationary combustion reported in 2006 IPCC guidelines for national GHG inventories (IPCC, 2006). The estimated CO_2 emissions from electricity generation used in the study is therefore slightly different and updated compared to values reported in IEA (2007a, d) which are based on 1996 IPCC emission factors. The fraction of carbon oxidized for each energy sources used in CO_2 emissions calculation are based on IEA (2007d). The values of fraction of carbon oxidized used for estimating net CO_2 emissions in Eq. (1) are 0.98 for coal and coal products, 0.99 for oil and oil products, 0.995 for natural gas and 0.99 for peat.

5. Results and discussion

In this section, results of factors affecting the changes in CO_2 emissions from electricity generation are discussed for APP countries and selected world regions both on temporal and period-wise basis over the historical (1990–2005) period. In addition, an analysis of future period (2005–2030) is presented for selected four APP countries.

Fig. 2 presents decomposition factors of CO_2 emissions changes in electricity generation sector with respect to the Kyoto base year 1990 for APP countries and OECD Europe as a whole. There are significant differences between these countries in terms of both magnitude and direction of factors affecting the changes in emissions. In general, there is an upward trend in *PDN* effect in all APP countries and this effect is responsible for most of the rise in emissions. With the exception of Australia and India, there is a downward trend in *EIN* effect in five countries reflecting the improvement in average efficiency of fossil-fuel-fired power plants during the 1990–2005 period. The trend of *GMX* effect however is mixed across the countries.

Among three effects, the *PDN* effect is the most dominant factor affecting rise in emissions in APP countries and OECD Europe average. At the country level, two emerging economies of the world—China and India—and the US show the largest total emissions increase while Canada shows the lowest emissions increase over the 1990–2005 period attributed mainly due to *PDN* effect. In contrast, the contribution from *GMX* effect remains about the same in Australia, China and India while its contribution is increasing in Canada, Japan, Korea and the US towards the later years of this period. In the case of *EIN* effect, five countries, excluding Australia and India, show decrease in emissions. It is noteworthy that in OECD Europe as a whole, both *GMX* and *EIN* effects are responsible for declining emissions while *PDN* effect is responsible for increasing emissions leading to overall decrease in total emissions during the same period.

Table 2 presents the result of aggregate decomposition analysis for the period 1990–2005. The *PDN* effect is the major factor for increase in emissions in APP countries accounting for over 95% of total increase. More specifically, the *PDN* effect accounted more than the total emissions increase in Canada, China and Japan between 1990 and 2005. The *GMX* effect is the second major factor of total emissions increase ranging from high of 138 Mt increase in the US to low of 6 Mt increase in Canada. With the exception of Australia and India, the *EIN* effect is the major factor responsible for decrease in emissions in five countries, ranging from 120 Mt decrease in China to 8 Mt decrease in Canada. In percentage terms, the *EIN* effect accounted for double digits decrease in four countries (Canada, Japan, Korea and the US).

Table 3 presents results of two sub-periods decomposition analysis, namely 1990–1997 and 1997–2005. These two periods are chosen to reflect the significant changes that took place in the Asian economy during and after the Asian financial crisis 1997–1998. During the first period (1990–1997), the *PDN* effect is the major factor accounting for 90% of total emissions increase followed by the *GMX* effect which accounted for 11% of the total increase. In contrast, the *EIN* effect accounted for 1% decrease during the same period.

At the country level, three countries (China, India and the US) together accounted for more than 86% of the total APP emissions increase with *PDN* effect being the major contributing factor. Except for Canada and Japan, the *GMX* effect is responsible for total emissions increase in five countries. This increase in emissions due to *GMX* effect in five countries is because of increase in coal share which is higher relative to decrease in oil share (Fig. 3a). In Canada and Japan, decrease in oil share is higher relative to increase in coal share and the substitution from oil to

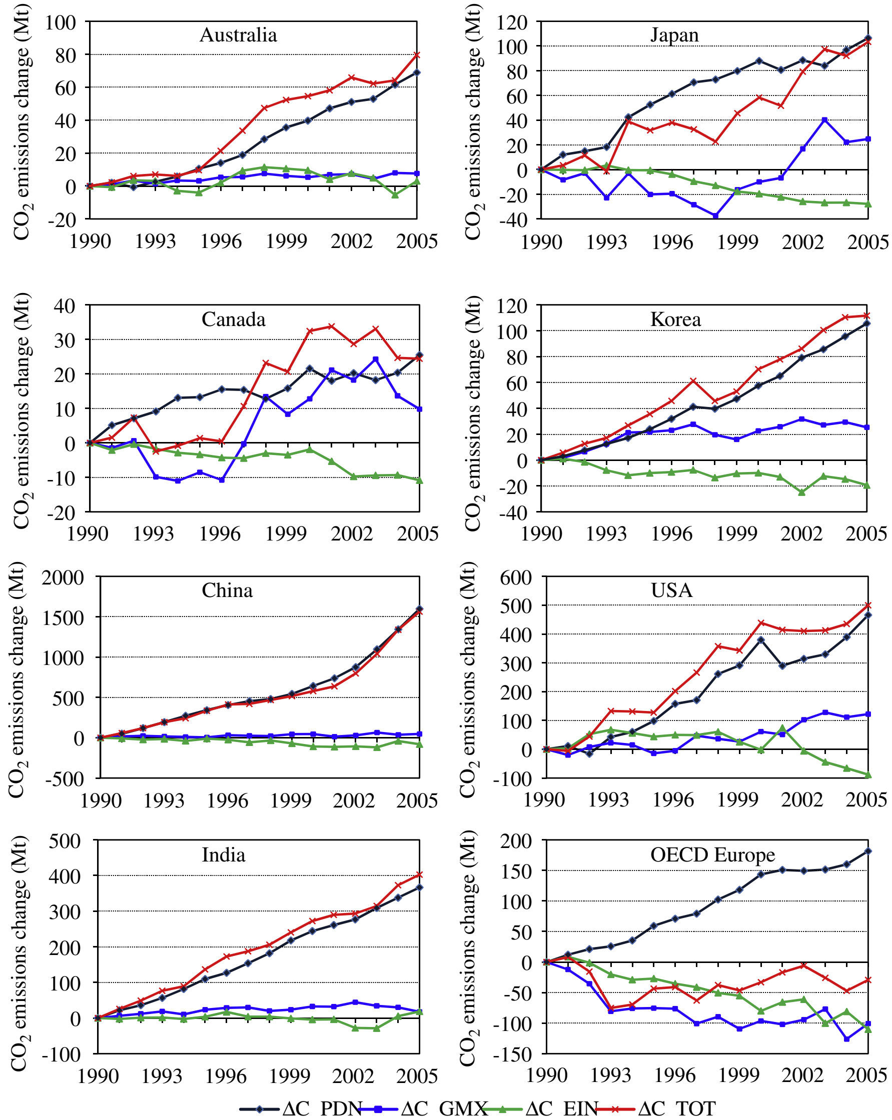

Fig. 2. Decomposition factors of CO_2 emissions changes in electricity generation sector compared to 1990 emissions level for seven APP countries and OECD Europe as a whole for the period 1990–2005.

Table 2
Decomposition factors of CO_2 emissions changes in electricity generation sector for the period 1990–2005 (Mt CO_2)

Country/region	Production effect	Electricity generation mix effect				Energy intensity effect				Total effect
		Coal	Oil	Gas	Sum	Coal	Oil	Gas	Sum	
Australia	66	14.1	−2.3	−1.3	11	−0.4	0.1	3.0	3	80
Canada	26	1.6	−0.1	4.9	6	−7.6	−1.2	1.1	−8	24
China	1605	127.5	−50.1	0.2	78	−119.4	−0.2	0.0	−120	1563
India	355	13.4	5.2	15.9	35	26.6	−4.7	−8.8	13	403
Japan	106	113.1	−93.7	5.7	25	−15.8	−6.7	−4.7	−27	103
Korea	88	59.4	−13.7	3.5	49	−20.8	−2.2	−2.6	−26	112
USA	415	62.6	−19.1	94.6	138	7.9	−20.1	−41.0	−53	500
APP total	2661	391.6	−173.8	123.6	341	−129.4	−34.9	−53.0	−217	2785
Developing Asia	297	52.9	9.3	−1.4	61	−5.9	−27.5	0.0	−33	324
OECD Europe	183	−182.5	−50.5	121.1	−112	−56.3	−21.3	−23.1	−101	−29

carbon-neutral nuclear, cleaner fuels and renewables energy sources lead to overall decrease in total emissions from *GMX* effect. For example, the share of oil in electricity generation in Japan decreased from 28% to 16% while the share of nuclear increased from 31% to 39% between 1990 and 1997. In Canada, oil is substituted by natural gas and renewables (mainly hydro

Table 3
Decomposition factors of CO_2 emissions changes in electricity generation sector for two sub-periods: 1990–1997 and 1997–2005 (Mt CO_2)

Country/region	1990–1997				1997–2005			
	ΔC_{PDN}	ΔC_{GMX}	ΔC_{EIN}	ΔC_{TOT}	ΔC_{PDN}	ΔC_{GMX}	ΔC_{EIN}	ΔC_{TOT}
Australia	19	5	9	34	50	4	−8	46
Canada	16	−1	−5	11	9	9	−4	14
China	445	29	−54	419	1158	28	−43	1144
India	152	32	3	188	212	−7	10	215
Japan	69	−27	−9	33	35	55	−19	71
Korea	41	31	−10	61	63	2	−15	50
USA	169	47	51	266	273	75	−115	234
APP total	911	116	−15	1011	1801	165	−193	1773
Developing Asia	137	32	−20	150	173	9	−7	175
OECD Europe	80	−103	−39	−63	101	−1	−67	34

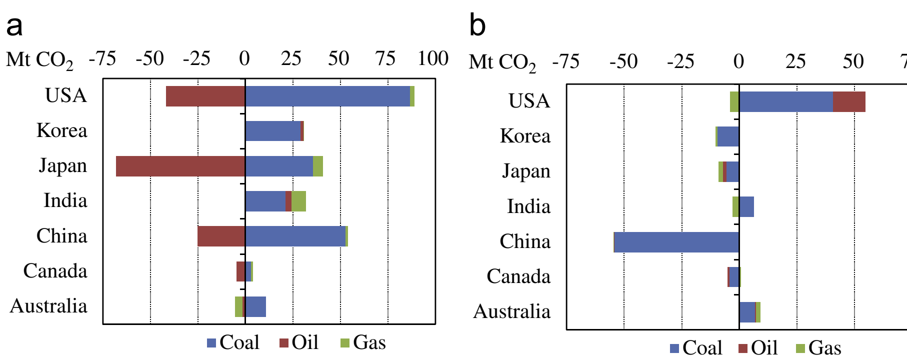

Fig. 3. Changes in CO_2 emissions from electricity generation mix and energy generation intensity effects by fuel type in APP countries for the period 1990–1997. (a) Electricity generation mix effect and (b) energy generation intensity effect.

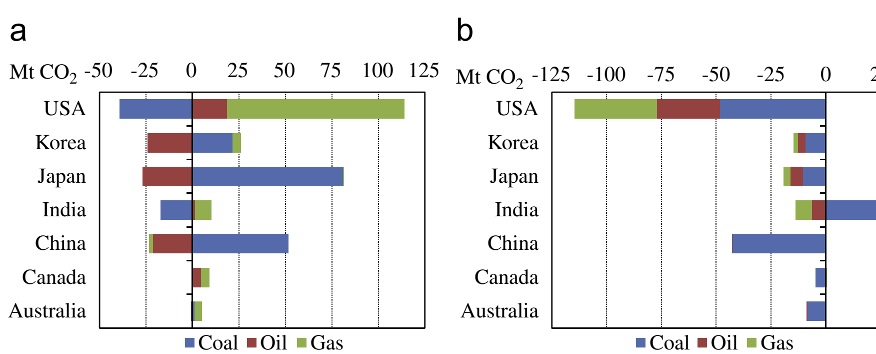

Fig. 4. Changes in CO_2 emissions from electricity generation mix and energy generation intensity effects by fuel type in APP countries for the period 1997–2005. (a) Electricity generation mix effect and (b) energy generation intensity effect.

electricity) during the same period. As for the *EIN* effect, it accounted for decrease in emissions in Canada, Japan, Korea and China while it accounted for an increase in emissions in Australia, the US and India. At the fuel level, the *EIN* effect of all fossil fuels (coal, oil and gas) accounted for decrease in emissions in Japan, while it is coal and oil in China and Canada, oil and gas in India, coal and gas in Korea and gas in the US (Fig. 3b). However, in Australia, the *EIN* effect of all fossil fuels accounted for increase in emissions and in India, only coal accounted for increase in emissions.

There is a significant change in factors affecting changes in CO_2 emissions in APP countries both in terms of magnitude and direction during the second period 1997–2005. The *PDN* effect is the major factor which would have exceeded the total emissions increase by 109% in the absence of *EIN* effect (Table 3). This increase however is offset by 11% decrease from *EIN* effect and 9% increase from *GMX* effect. Three countries (China, India and the US) combined are responsible for 90% of total APP countries emissions increase. Except for India, the *GMX* effect in six countries is responsible for increase in emissions (Fig. 4a). Substitution of oil by coal and gas contributed to positive *GMX* effect in Australia, Japan and Korea. Likewise, substitution of coal by oil and gas in Canada, substitution of oil and gas by coal in China and substitution of coal by oil and gas in the US contributed to positive *GMX* effect. In the case of India, substitution of coal by oil and gas contributed to overall negative *GMX* effect. As for the *EIN* effect, six countries (excluding India) show decrease in emissions ranging from 49% in the US to 4% in China. Increase in average efficiency of coal, oil and gas power plants in these six countries attributed to decrease in emissions due to *EIN* effect

Table 4
Decomposition factors of CO_2 emissions changes in power sector[a] for the period 2005–2030 (Mt CO_2)

Country/region	Reference case				Alternative case			
	ΔC_{PDN}	ΔC_{GMX}	ΔC_{EIN}	ΔC_{TOT}	ΔC_{PDN}	ΔC_{GMX}	ΔC_{EIN}	ΔC_{TOT}
China	4787	−31	−1154	3603	3649	−622	−869	2158
India	1589	39	−402	1227	1096	−185	−296	615
Japan	113	−91	−4	18	66	−175	−9	−117
USA	895	−25	−385	486	594	−339	−325	−70
Developing Asia	678	20	−133	565	395	−129	−94	171
OECD Europe	475	−6	−170	299	190	−560	−9	−379

[a] Includes both electricity and heat generation.

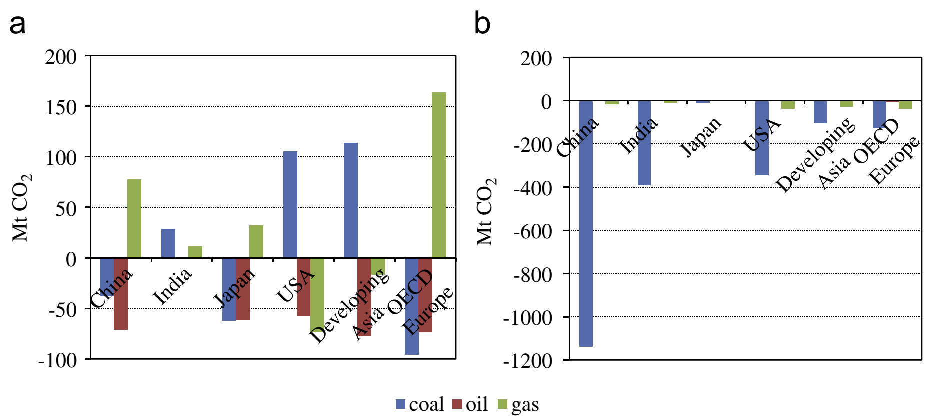

Fig. 5. Changes in CO_2 emissions from electricity generation mix and energy generation intensity effects by fuel type in the reference case for the period 2005–2030. (a) Electricity generation mix effect and (b) energy generation intensity effect.

(Fig. 4b). The reason for positive *EIN* effect in India is because of decrease in average efficiency of coal-fired power plants during this period. For example, the average efficiency of coal-fired power plants in India decreased from 27.8% in 1997 to 26.5% in 2005.

Table 4 presents the factors affecting the changes in CO_2 emissions from power generation for four major APP countries (China, India, Japan and the US) and the selected world regions for the period 2005–2030 under two scenarios, namely, the reference case and the alternative policy case taken from IEA (2007a).[3] Broadly, the reference case is a baseline vision of how global energy markets are likely to evolve if there are no new energy-policy interventions by governments. However, it takes account of those government policies and measures that had already been adopted by mid-2007, e.g., measures to boost biofuels in the US, new measures to promote renewables in Japan and the mandatory phase-out of incandescent light bulbs in Australia. In contrast, the alternative policy case analyses the impact on global energy markets of a package of additional measures to address energy-security and climate-change concerns. Some of these measures include more rigorous action to promote renewables, including biofuels, more stringent energy-efficiency standards and more ambitious plans for nuclear power.

In the reference case, two major Asian developing country economies (China and India) are estimated to be by far the biggest contributor to incremental power sector emissions overtaking the US by 2030. These two Asian countries are projected as the world's two biggest power sector CO_2 emitters by 2030. Between 2005 and 2030, the total power sector CO_2 emissions is estimated to increase by 183% in India, followed by China (146%), the US (20%) and Japan (4%). The *PDN* effect would contribute more than the total emissions increase in the absence of *GMX* and *EIN* effects in all four countries during this period. The *GMX* effect would be responsible for decrease in total emissions in China, Japan, the US and OECD Europe as a whole. In particular, the *GMX* effect would contribute more than 500% decrease in total emissions in Japan between 2005 and 2030. Substitution of oil by coal and gas contributed to positive GMX effect in Australia, Japan and Korea. This decrease in emissions in Japan is mainly due to decrease in shares of coal and oil in electricity generation (Fig. 5a) and increase of carbon-neutral nuclear energy source in Japan. Likewise, decrease in coal and oil shares in total fossil-fuel-based electricity generation is more than the increase in gas share in China and OECD Europe as a whole and decrease in oil and gas shares is more than increase in coal share in the US contributed to negative GMX effect (Fig. 5a). In contrast, the *GMX* effect would be responsible for increase in emissions in India (3%) and the DA as a whole (4%) over the same period. In the case of *EIN* effect, it would be responsible for decrease in emissions in all four countries including DA and OECD Europe as a whole (Fig. 5b). This decrease in emissions ranges from high of 79% in the US to low of 24% in Japan and DA as a whole.

In the alternative policy case, developing Asian countries (China, India and DA) show increase in total emissions while industrialized nations (Japan and the US) including OECD Europe as a whole show decrease in total emissions during the 2005–2030 period. For example, the total power sector emissions between 2005 and 2030 is estimated to increase by 92%, 88% and 34% in India, China and DA as a whole, respectively, while this total emissions is estimated to decrease by 28%, 26% and 3% in

[3] Due to data constraints, factors decomposition analysis for period 2005–2030 is limited to power generation—which includes both electricity and heat generation—for four major APP countries and selected world regions as reported in IEA (2007a).

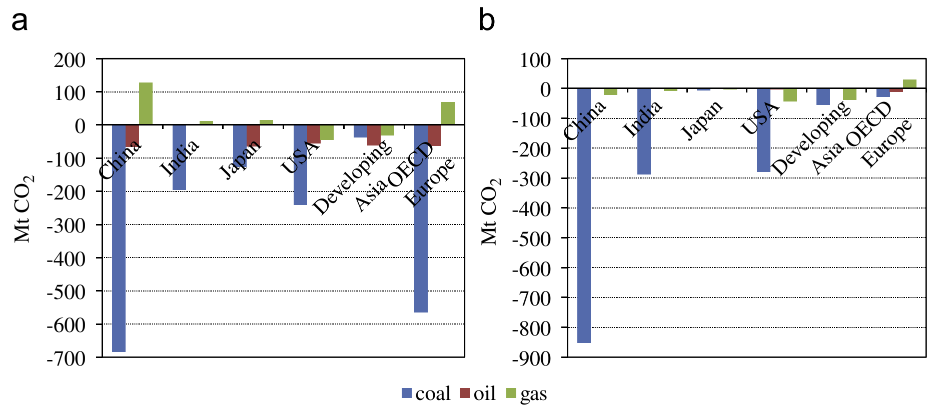

Fig. 6. Changes in CO_2 emissions from electricity generation mix and energy generation intensity effects by fuel type in the alternative policy case for the period 2005–2030. (a) Electricity generation mix effect and (b) energy generation intensity effect.

Table 5
Efficiency improvement of fossil-fuel-fired power plants and its co-benefits in APP countries in year 2005

Country	Efficiency improvement (%)			Energy saved[a] (Mtoe)			Avoided CO_2 emissions (Mt CO_2)			Amount saved[a] (million US$)		
	Coal	Oil	Gas	Coal	Oil	Gas	Coal	Oil	Gas	Coal	Oil	Gas
Australia	6.8	8.7	21.7	3.16	0.04	1.35	12.3	0.1	3.1	596	22	889
Canada	3.6	6.6	10.2	0.86	0.27	0.51	3.3	0.9	1.2	118	163	325
China	10.1	13.8	13.6	53.21	2.10	0.36	206.6	6.7	0.8	7052	1224	243
India	15.7	15.2	10.6	24.40	1.25	1.36	94.7	4.0	3.2	4545	757	818
Japan	–	–	7.2	–	–	3.18	–	–	7.4	–	–	2132
Korea	2.9	4.5	–	0.85	0.18	–	3.3	0.6	–	203	111	–
USA	5.4	11.1	6.7	26.29	3.06	7.26	102.1	9.8	17.0	3584	1857	4643

[a] Calorific values and energy prices used in estimating energy saved and amount saved are based on IEA (2005), IEA (2006b), and IEA (2007a–c).

OECD Europe as a whole, Japan and the US, respectively. The *PDN* effect would be the main factor accounting for increase in total emissions in all four countries including both OECD Europe and DA as a whole while both *GMX* and *EIN* effects would be responsible for decrease in total emissions during the same period. In general, substitution of coal and oil by natural gas (Fig. 6a) and increase shares of nuclear and renewable energy sources in total electricity generation together with the application of advance clean coal technologies (Fig. 6b) is responsible for decrease in emissions due to *GMX* and *EIN* effects.

Overall, the decomposition analysis finds that efficiency improvement of fossil-fuel-fired power plants—although varies across APP countries—is the main factor for decrease in total emissions for the period 1990–2005. This efficiency gap across APP countries suggests that there is a potential for decrease in fossil fuels demand and corresponding CO_2 emissions reduction as well as financial benefits from mutual cooperation. Specifically, coal-fired electricity generation is of particular interest from efficiency improvement point of view. This is because coal constitutes a major share in total fossil-fuel-fired power plants in these countries and this share is expected to remain strong for next 25 years. Furthermore, both higher oil and gas prices in recent years and energy-security concerns are making coal more competitive and attractive as a fuel for electricity generation in most of these countries. In addition, advance clean coal technologies incorporating CO_2 capture and storage (CCS) offer promising CO_2 mitigation measure from coal use in electricity generation in the future.

Table 5 presents the potential of energy savings, CO_2 emissions reduction and financial benefits if all countries produce electricity at the highest efficiencies observed for the included countries. For example, in 2005, average coal-fired power plant efficiency improvement in the range of 2.9% (Korea) to 15.7% (India) could have been achieved if the best practice technology in Japan is used. This efficiency improvement translates into reduction of total coal requirements by 109 Mtoe of which 37% is in India and 24% is in China. The equivalent CO_2 emissions avoided would have ranges from 3.3 Mt in Canada to about 207 Mt in China. More important is the total amount saved from reduced energy demand which would have been more than 16 billion US$ in 2005 alone for all countries combined. In the case of oil-fired power plants, average efficiency improvement is in the range of 4.5% in Korea to 15% in India if using the best practice technology in Japan. This corresponds to about 10 Mt of CO_2 emissions avoided and about 4 billion US$ savings for all countries combined. Likewise, average natural gas electricity generation efficiency improvement if using the best practice technology in Korea ranges from 6.7% in the US to about 21% in Australia. This efficiency improvement corresponds to 13% reduction in natural gas requirements, 17 Mt of CO_2 emissions avoided and about 9 billion US$ savings in 2005 for all countries combined. The potential for energy savings and emissions reduction would be even higher if using commercially available best practice plants with efficiencies as high as average 47% for conventional coal-fired power plants (ultra-supercritical units) and 60% for natural gas-fired combined cycle plants (IEA, 2006c).

6. Conclusions

APP countries together share significant portion of global economy, energy use and CO_2 emissions. In 2005, these countries combined is responsible for more than half of the world's energy-related total CO_2 emissions and more than two-thirds of world's CO_2 emissions from electricity generation, with China, the US, India and Japan dominating the totals. Between 1990 and 2005, total CO_2 emissions from electricity generation almost doubled in APP countries combined in contrary to decrease in OECD Europe as a whole.

The decomposition factors of CO_2 emissions changes from electricity generation in APP countries presented in this study identifies production effect as the major factor responsible for increase in emissions followed by electricity generation structural effect. In contrary, the energy intensity of electricity generation effect is a key factor responsible for decrease in emissions. However, the degree of influence of these factors affecting changes in emissions varies from country to country. The sub-period decomposition analysis shows that the Asian financial crisis has no role in slowing down the rising emissions in these countries. In fact, the relative magnitude of total emissions increase is larger in post-Asian financial crisis period (1997–2005) than in pre-crisis period (1990–1997) except in Korea and the US. In these two countries, efficiency improvement of fossil-fuel-fired power plants is responsible in containing the rise in total emissions. In next 25 years (2005–2030), reference case results find that production effect remains the key factor responsible for increase in emissions in all four major countries (China, India, Japan and the US). But, unlike in the past, electricity generation structural effect contributes significant decrease in emissions in China, Japan and the US. In addition, the energy intensity of electricity generation effect is responsible for decrease in emissions in all four countries during this period.

Over the period 1990–2005, the share of oil use has decreased while shares of coal and gas use have increased in total fossil-fuel-fired electricity generation. Coal use in APP countries remains the dominant fuel in total fossil-fuel-fired electricity generation and it is by far the biggest contributor to incremental emissions between 1990 and 2005. Furthermore, both higher oil and gas prices in recent years and energy-security concerns are making coal even more competitive and attractive as a fuel for electricity generation in these countries. With limited role of renewables and strong public opposition to carbon-neutral nuclear power energy, the CO_2 emissions from electricity generation in APP countries will continue to increase in future in the absence of any mitigation measures.

The results find that the largest users of fossil fuels for electricity generation are also among the least efficient. Therefore, there is a potential of efficiency improvement of fossil-fuel-fired power plants in APP countries and its associated co-benefits, including reduced energy demand and CO_2 emissions, and financial benefits. The other promising measures in containing the growth of CO_2 emissions from electricity generation may include but not limited to promotion of renewable portfolio standards (RPS) by the governments and application of advance clean coal technologies incorporating CO_2 capture and storage (CCS). However, new policies and financing schemes are needed to encourage these measures.

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